

1. First principles: what market-risk capital is trying to achieve

A trading book can gain or lose value because market prices move. A bond changes value when rates or credit spreads move. An option changes value when the underlying price, volatility, rates or correlations move. A credit instrument can jump in value when an issuer defaults. A structured product can contain risks that are difficult to represent with a small set of smooth sensitivities.

Regulatory market-risk capital is a solvency buffer against these adverse movements. It is not a forecast of tomorrow's P&L and it is not a replacement for limits, stress testing, valuation controls or risk management. Its job is to provide a conservative, comparable measure of loss-absorbing capacity for market risk.

Before FRTB, internal-model capital was built primarily around Value-at-Risk (VaR), stressed VaR and incremental/default-related measures. The financial crisis exposed several weaknesses: tail losses beyond the VaR quantile could be very different across portfolios, liquidity could disappear in stress, trading-book boundaries could be exploited, and internal-model approval could apply too broadly relative to actual model performance.

FRTB addresses these issues through four design choices:

1. **A more risk-sensitive Standardised Approach.** The SA decomposes market risk into delta, vega, curvature, default and residual components and prescribes risk weights and correlations.
2. **Expected Shortfall for modelled market risk.** IMA uses a 97.5% one-tailed ES and explicitly adjusts for risk-factor liquidity horizons.
3. **Desk-level model approval.** A bank does not receive one blanket IMA permission for the whole trading book. Model eligibility is assessed at trading-desk level.
4. **Data and performance gates.** Risk-factor modellability, P&L attribution and backtesting determine which risks and desks may remain in the internal model.

A useful mental model is: **SA asks “what sensitivities/default/residual risks do you have under prescribed supervisory shocks?”** IMA asks “how large are tail losses under an approved model, after forcing stress, liquidity, modellability and model-performance constraints?”

1.1 Capital versus RWA

In the Basel market-risk framework, once a market-risk capital requirement K is determined, risk-weighted assets are:

$$RWA_{MR} = 12.5 K.$$

The factor 12.5 is 1/8%. It converts an 8%-of-RWA minimum-capital convention into an RWA equivalent. If a market-risk charge is 80 million, its RWA equivalent is 1,000 million.

1.2 Sensitivity, shock and loss

Suppose the value of a position V depends on a market variable x . A small change is approximately

$$\Delta V \approx \frac{\partial V}{\partial x} \Delta x.$$

The derivative is the first-order sensitivity. FRTB SA starts from this familiar idea, then imposes standardised shocks through risk weights and constrains diversification through regulatory correlations.

For nonlinear instruments, a second-order approximation is

$$\Delta V \approx \Delta \Delta S + \frac{1}{2} \Gamma (\Delta S)^2 + \text{other terms}.$$

Delta capital primarily reflects the first term. Curvature exists because the second and higher-order terms can produce material losses under large shocks.

2. FRTB architecture

2.1 Standardised Approach

Under the Basel SA structure:

$$K_{SA} = K_{SBM} + K_{DRC} + K_{RRAO}.$$

- **SBM**: sensitivities-based method for delta, vega and curvature.
- **DRC**: default risk capital for jump-to-default risk.
- **RRAO**: residual risk add-on for specified risks inadequately captured by SBM and DRC.

The calculation is deliberately modular. A convertible bond can generate GIRR, CSR, equity and FX sensitivities in SBM, default risk in DRC, and potentially residual-risk treatment depending on its structure. The modules are not alternative ways to charge the same trade; a trade can be in several modules because it carries several distinct risk types.

2.2 Internal Models Approach

IMA has several layers:

1. trading-desk eligibility and supervisory approval;
2. risk-factor definition and mapping;
3. Risk Factor Eligibility Test (RFET) to separate modellable risk factors from NMRFs;
4. liquidity-adjusted 97.5% Expected Shortfall for modellable factors;
5. stressed calibration using a reduced factor set;
6. stress-scenario capital for NMRFs;
7. P&L Attribution (PLA) and desk-level backtesting;
8. an internal-model default-risk charge for eligible default-risk portfolios;
9. capital aggregation, including standardised capital for desks that are not IMA-eligible and an amber-zone surcharge where applicable.

2.3 Why SA still matters to an IMA bank

SA is not made irrelevant by IMA. It remains important for:

- desks without IMA approval;
- desks that lose IMA eligibility;
- capital comparisons and the PLA amber-zone surcharge;
- output-floor and jurisdiction-specific uses;
- implementation benchmarking and model-risk diagnostics.

This is why mature FRTB architecture generally maintains a robust SA engine even when the strategic objective is IMA.

3. The Sensitivities-Based Method: the universal calculation pattern

SBM defines seven risk classes:

1. General Interest Rate Risk (GIRR)
2. Credit Spread Risk (CSR) — non-securitisations
3. CSR — securitisations, non-correlation trading portfolio (non-CTP)
4. CSR — securitisations, correlation trading portfolio (CTP)
5. Equity
6. Commodity
7. Foreign Exchange (FX)

For delta and vega, the logic is the same across risk classes.

3.1 3.1 Step 1 — calculate and net sensitivities

Let s_k be the net sensitivity to regulatory risk factor k . All instrument-level sensitivities mapped to the same risk factor are algebraically netted before risk weighting.

A model or risk engine can produce thousands or millions of trade-level sensitivities. Regulatory mapping turns them into a smaller set of risk-factor coordinates: currency/curve/tenor for GIRR, issuer/sector/credit quality/tenor for CSR, market-cap/economy/sector for equity, commodity type and tenor for commodities, and reporting-currency exchange-rate factors for FX.

3.2 3.2 Step 2 — risk-weight each sensitivity

$$WS_k = RW_k s_k.$$

The risk weight RW_k represents the prescribed severity of the supervisory shock for that risk factor. Two equal raw sensitivities can have different weighted sensitivities if their liquidity or risk classification differs.

3.3 3.3 Step 3 — aggregate within a bucket

For a bucket b :

$$K_b = \sqrt{\max \left(0, \sum_k WS_k^2 + \sum_{k \neq l} \rho_{kl} WS_k WS_l \right)}.$$

Here ρ_{kl} is the prescribed within-bucket correlation between risk factors k and l .

If the weighted sensitivities are 100 and -100 and $\rho = 1$, the bucket charge can cancel completely. If $\rho = 0.5$, the charge is

$$\sqrt{100^2 + (-100)^2 + 2(0.5)(100)(-100)} = 100.$$

The economic message is direct: opposite signs are not enough for full capital relief. The rule asks how reliable the hedge is under stress.

3.4 3.4 Step 4 — aggregate across buckets

Define the signed bucket sum

$$S_b = \sum_{k \in b} WS_k,$$

with the regulatory bounding treatment used in the cross-bucket formula where required. The risk-class charge has the form

$$K = \sqrt{\max \left(0, \sum_b K_b^2 + \sum_{b \neq c} \gamma_{bc} S_b S_c \right)},$$

where γ_{bc} is the prescribed cross-bucket correlation.

3.5 3.5 Step 5 — three correlation scenarios

FRTB calculates SBM under three correlation regimes.

- **Medium:** prescribed ρ and γ .
- **High:** multiply each relevant correlation by 1.25 and cap at 100%.
- **Low:** transform a correlation x as

$$x_{low} = \max(2x - 100\%, 75\%x).$$

The final SBM charge uses the largest result across the three correlation scenarios. This prevents a portfolio from relying on a single diversification assumption.

Worked example

Suppose one bucket contains $WS_1 = 80$ and $WS_2 = -45$ with medium correlation 60

$$K_b = \sqrt{80^2 + 45^2 + 2(0.6)(80)(-45)} \approx 64.93.$$

Under the high scenario, correlation becomes 75

4. Delta risk by risk class

4.1 GIRR: interest-rate curves, inflation and cross-currency basis

A GIRR bucket is a currency. Basel delta risk factors for a risk-free yield curve use tenors:

Tenor	0.25y	0.5y	1y	2y	3y	5y	10y	15y	20y	30y
Basel RW	1.7%	1.7%	1.6%	1.3%	1.2%	1.1%	1.1%	1.1%	1.1%	1.1%

Inflation and cross-currency-basis risk factors carry a Basel risk weight of 1.6%.

For EUR, USD, GBP, AUD, JPY, SEK, CAD and the bank's domestic reporting currency, Basel permits the above GIRR risk weights to be divided by $\sqrt{2}$ at the bank's discretion under MAR21.44. This is a calibration rule, not a generic "liquid currency" judgment.

4.1.1 Curve dimension matters

GIRR is not merely "one DV01 per currency." Different regulatory curves can coexist in the same currency bucket. OIS, market-implied swap curves and other prescribed curve dimensions create basis risk. Within the same tenor but different curves, Basel uses a very high correlation of 99.90%, which still leaves a small basis charge for offsetting positions.

Inflation has its own risk factor treatment, and cross-currency basis is separately identified. Onshore/offshore and curve construction choices can therefore be capital relevant.

4.1.2 Worked GIRR example

Assume a USD single-curve portfolio with sensitivities (in the reporting currency):

- 1y: +600
- 5y: -350
- 10y: +250

Using the $\sqrt{2}$ treatment, the risk weights are $1.6\%/\sqrt{2}$, $1.1\%/\sqrt{2}$, $1.1\%/\sqrt{2}$. Weighted sensitivities are therefore approximately 6.79, -2.72 and 1.94. The same-curve Basel tenor correlations for 1y–5y, 1y–10y and 5y–10y are 88.7%, 76.3% and 97.0% respectively. Insert them into the quadratic aggregation and then repeat under high/low correlation scenarios. The offline lab performs this calculation interactively.

4.2 CSR non-securitisations

CSR delta measures changes in value from credit-spread movements separate from default. The risk-factor dimensions include issuer/curve and prescribed tenors of 0.5y, 1y, 3y, 5y and 10y. Bucket assignment is driven by credit quality and sector.